CO2-AT-2

Assessment tool-2

1. Morphological Processing

```
words = ["analyzing", "analysis", "analytical"]

morphology = {
    "analyzing": ["analyze", "-ing", "Inflectional"],
    "analysis": ["analysis", "None", "Lexical form"],
    "analytical": ["analysis", "-ical", "Derivational"]
}

print("Word\t\tRoot\t\tSuffix\t\tType")
print("-" * 65)

for word in words:
    root, suffix, word_type = morphology[word]
    print(f"{word}\t{root}\t{suffix}\t\t{word_type}")

    print("Normalized form: ANALYS")
```

```
Word            Root            Suffix          Type
-----------------------------------------------------------------
analyzing       analyze -ing            Inflectional
Normalized form: ANALYS
analysis        analysis        None            Lexical form
Normalized form: ANALYS
analytical      analysis        -ical           Derivational
Normalized form: ANALYS
```

---

2. Morphological Parsing

```
words = {
    "disagree": ["dis-", "agree", "None", "Not agree"],
    "agreement": ["None", "agree", "-ment", "State of agreeing"],
    "agreeable": ["None", "agree", "-able", "Pleasant or acceptable"]
}

print("Word\t\tPrefix\tRoot\tSuffix\tMeaning")
print("-" * 75)

for word, data in words.items():
    prefix, root, suffix, meaning = data
    print(f"{word}\t{prefix}\t{root}\t{suffix}\t{meaning}")
    print("Normalized form:", root)
```

```
---------------------------- RESTART: C:/Users/Ambuselvi/rty.py ---------
Word            Prefix  Root    Suffix  Meaning
------------------------------------------------------------------------
disagree        dis-    agree   None    Not agree
Normalized form: agree
agreement       None    agree   -ment   State of agreeing
Normalized form: agree
agreeable       None    agree   -able   Pleasant or acceptable
Normalized form: agree
```

---

3. Morphological Normalization

```
words = {
    "govern": ["govern", "None", "Base form"],
    "government": ["govern", "-ment", "First derivational level"],
    "governance": ["govern", "-ance", "First derivational level"]
}

print("Word\t\tRoot\tSuffix\tLevel")
print("-" * 70)

for word, data in words.items():
    root, suffix, level = data
    print(f"{word}\t{root}\t{suffix}\t{level}")
    print("Normalized form:", root)
```

```
---------------------------- RESTART: C:/Users/Ambuselvi/rty
Word            Root    Suffix  Level
----------------------------------------------------------
govern  govern  None    Base form
Normalized form: govern
government      govern  -ment   First derivational level
Normalized form: govern
governance      govern  -ance   First derivational level
Normalized form: govern
```

---

4. Morphological Parsing and Normalization

```
words = {
    "activate": ["None", "active", "-ate", "Verb"],
    "activation": ["None", "activate", "-ion", "Noun"],
    "reactivation": ["re-", "activate", "-ion", "Noun"]
}

print("Word\t\tPrefix\tRoot\t\tSuffix\tClass")
print("-" * 65)

for word, data in words.items():
    prefix, root, suffix, word_class = data
    print(f"{word}\t{prefix}\t{root}\t{suffix}\t{word_class}")
    print("Normalized form: activate")
```

```
Word            Prefix  Root            Suffix  Class
-----------------------------------------------------------------
activate        None    active  -ate    Verb
Normalized form: activate
activation      None    activate        -ion    Noun
Normalized form: activate
reactivation    re-     activate        -ion    Noun
Normalized form: activate
```

---

5. Inflectional Morphology

```
words = {
    "create": ["None", "Base form"],
    "creates": ["-s", "Third-person singular present tense"],
    "creating": ["-ing", "Present participle"]
}

print("Word\t\tSuffix\tCategory\t\t\tNormalized")
print("-" * 80)

for word, data in words.items():
    suffix, category = data
    print(f"{word}\t{suffix}\t{category}\tcreate")
```

```
Word            Suffix  Category                        Normalized
--------------------------------------------------------------------------------
create  None    Base form       create
creates -s      Third-person singular present tense     create
creating        -ing    Present participle      create
```